Full-table FAIL root-cause report - compose_eos v3.3.0

Date: 2026-11-06
Input: `eos_data_normalized/`
Matched matrix: 13 crusts x 5 cores x 6 bridge methods = 390 tables
Independent reparse/recheck: 390/390 successful

Verdict

The 156 full-table FAIL results are **not cross-EOS matching conflicts and are not produced by the interpolation or checker implementation**. Every one is classified `SOURCE_CORE_ACAUSALITY_PRESERVED`. They split exactly into 78 combinations using `APR_APR` and 78 using `RG_KDE0v1`. The other 234 combinations pass all hard gates. The causal source tails in each of the 156 failed matched tables are numerically identical to the corresponding core input after the declared transition point. The bridge did not create the superluminal segment.

Classification	Count	Meaning
PASS	234	The full matched EOS passes all hard gates
SOURCE_CORE_ACAUSALITY_PRESERVED	156	Source core already contains c_s^2 > 1, and the matched high-density tail preserves it
MATCHING_OR_NUMERICAL_CANDIDATE	0	No residual failure was attributed to the bridge after independent readback
NON_C10_GATE_FAILURE / AUDIT_READ_ERROR	0	No such case was observed in the 390-table audit

Independent derivative checks on the *source* core tables

The sound speed was recalculated from the source tables using three distinct derivative constructions:

$$c_s^2 = \frac{dP}{d\epsilon} = \frac{P}{\epsilon}\frac{d\ln P/d\ln n_B}{d\ln\epsilon/d\ln n_B}.$$

Core source	PCHIP max c_s^2	Natural cubic max c_s^2	Centered-FD max c_s^2	PCHIP first c_s^2 > 1 (fm^-3)
APR_APR	1.471111	1.474094	1.471111	0.880000
RG_KDE0v1	1.107147	1.109488	1.107147	1.442943
BBB_BHF2	0.857718	0.869686	0.857718	n/a
GM_GM1	0.756851	0.757183	0.756851	n/a
RG_SLY4	0.954303	0.955624	0.954303	n/a

For APR, the superluminal tail begins near `n_B = 0.88 fm^-3` and reaches approximately `c_s^2 = 1.47`. For KDE0v1 it begins near `1.442943 fm^-3` and reaches approximately `1.107`. All three derivative schemes reproduce the superluminal behavior. The numerical spread in the maximum is small compared with the excess above unity; this is not a one-scheme interpolation artifact.

Matched-tail identity test

For every failed directory, the audit located `nb_transition_right_fm3` and compared the remaining matched-core arrays with the source core arrays. All 156/156 failed cases preserve the source tail

in `n_B`, `P`, and `epsilon` within the declared tight floating-point tolerances. This establishes provenance of the failing segment, rather than merely correlating the failure with a source model name.

Physical, numerical, matching, and implementation attribution

- **Source/model physics:** The supplied APR and KDE0v1 tables, as represented over their full tabulated high-density ranges, violate relativistic causality. This says the *selected tabulated model/extrapolation* is acausal there. It does not establish that the true neutron-star EOS is acausal; the microscopic model may not be valid to those densities.
- **Numerics:** PCHIP, natural cubic spline and centered finite differences all identify the same superluminal tails. There is no evidence that the 156 FAIL count is generated by a derivative-scheme choice.
- **Crust-core matching:** The 156 failed tables have a matching-to-core tail equality proof. Cross-model saturation-property mismatch can still be a source of physical systematic uncertainty in the junction, but it is not the cause of these C10 failures.
- **Implementation:** The independent re-read and consistency re-evaluation reproduce the saved status. No code exception, malformed EOS, or metadata disagreement accounts for the 156 failures.

Optional causal completions are separate models

A maximally stiff or constant-sound-speed extension is not a correction uniquely implied by the microscopic table. It changes the high-density model and must be recorded as such.

- APR_APR with a CSS tail `c_s^2 = 0.95` matches at `n_B = 0.873272 fm^-3`; its full gate passes and PCHIP, cubic-spline and centered-FD maxima are approximately 0.9818, 0.9953 and 0.9809. This is a robust causal **alternative model**, not the original APR EOS.
- KDE0v1 with a CSS tail `c_s^2 = 0.80` passes the package finite-difference hard gate, but independent PCHIP and cubic-spline profiles still register approximately 1.00095 and 1.03225 near the join. It is therefore **diagnostic-only and not promoted**. The source onset is too close to the interpolation-sensitive join for this particular completion recipe to be called robust.

Conclusion

The correct resolution is to preserve the original 156 FAIL classifications, report their source provenance, and expose optional causal-completion variants with independent scheme validation. Reclassifying the original APR/KDE0v1 tables as PASS by clipping `c_s^2`, deleting their high-density points, or altering their tail would conceal rather than resolve the input-model limitation.